Please write clearly in block capitals.

Centre number MFS01 | | | | | Candidate number 2026 | | | |

Surname MASTER

Forename(s) FUNCTIONAL SKILLS

Candidate signature Master Functional Skills Date: 20/10/2026

Functional Skills Level 1
MATHEMATICS
Section A: Non-Calculator Paper
Section B: Calculator Paper

OFFICIAL MFS MODEL SOLUTION

60 / 60 Marks (100% Pass)

· Complete Stepped Working & Mark Codes

· Exam Board: Open Awards

Verified by Tutor Abdul · MFS

Paper A/B

Time allowed: 30 minutes / 1 hour 30 minutes

Materials

For this paper you must have:

- mathematical instruments.

You **must not** use a calculator for section A.
You may use a calculator for section B.

Instructions

- Use black ink or black ball-point pe
- Answer all questions.
- You must answer the questions in th around each page or on blank pages.
- Do all rough work in this book. Cros d.
- State the units of your answer wher

Information

- The marks for questions are shown in
- The maximum mark for section A is 15
- The maximum mark for this paper is 6
- You may ask for more answer paper, g These must be tagged securely to this

Advice

- In all calculations, show clearly how y

PASS FUNCTIONAL SKILLS

MASTER FUNCTIONAL SKILLS

ONLINE LEARNING PLATFORM & ACCREDITED COURSES

Pearson Edexcel & Open Awards Accredited Suite · Entry 3, Level 1 & Level 2

Pass Your Functional Skills Exams With Confidence

100% Pass Guarantee | Dyslexia-Friendly Learning | 1-to-1 Expert Tutor Support

Master Functional Skills provides an adaptive, dyslexia-friendly online learning platform tailored for adult learners, apprentices, and healthcare & vocational staff. Everything you need to revise, practice, and pass is at your fingertips.

1. Smart Diagnostic Initial Assessment

Calculate your exact working level and pinpoint knowledge gaps instantly

- **Exact Grade Calculation:** Adaptive assessments calculate your exact level across all topics.
- **Personalised Study Pathway:** Skip what you know; focus 100% on the topics needed to pass.
- **Real-Time Exam Readiness:** Predictive tracker confirms when you are 100% ready to book.

2. Comprehensive Video Lessons & Worked Solutions

Bite-sized visual explainers covering every single Maths & English curriculum standard

- **Step-by-Step Video Lessons:** Master percentages, algebra, perimeter, ratio, and reverse fractions.
- **Multi-Ink Handwriting Models:** Visual 3-colour methods showing working, carries, and examiner tips.
- **Tutor Shortcuts & Golden Rules:** Proven non-calculator mental arithmetic hacks and BIDMAS tricks.

3. Full Mock Exams & Real-Time Performance Analytics

Practice in exam conditions and watch your scores and confidence climb

- **Authentic Exam Papers:** Timed Section A & Section B papers formatted exactly like Pearson Edexcel.
- **Instant Topic Breakdown:** View percentage scores across Number, Measure/Shape, and Data.
- **Historical Score Tracking:** Monitor past attempts to ensure continuous improvement before exam day.

Ready to Master Your Functional Skills & Pass First Time?

Join thousands of healthcare staff, apprentices, and adult learners across the UK.

Instant 24/7 access on desktop, tablet, and mobile with no software to install.

START LEARNING TODAY AT

MasterFunctionalSkills.co.uk

SECTION A : Non-Calculator

Answer ALL questions.
Write your answers in the spaces provided.

© https://passfunctionalskills.co.uk/

1 Work out $19^2 - 3 \times 5 + 2$

[2 marks]

BIDMAS: $19^2 = 19 \times 19 = 361$; $3 \times 5 = 15$ [M1]

$361 - 15 + 2 = 346 + 2 = 348$ [A1]

Answer 348 ✓

2 Calculate $17.89 + 23.47$

[1 mark]

Align decimal points vertically: $0.89 + 0.47 = 1.36$; $17 + 23 + 1.36 = 41.36$

$17.89 + 23.47 = 41.36$ [1 mark]

Answer 41.36 ✓

3 Calculate $23.1 \div 1000$

[1 mark]

Divide by 1000 (move decimal point 3 places left):

$23.1 \div 1000 = 0.0231$ [1 mark]

Answer 0.0231 ✓

Turn over for next question

© https://passfunctionalskills.co.uk/

4 Jeffrey records the ages of some people that attend his gym.

26	15	28	18	19	43	53	61	48
28	34	42	23	25	22	31	20	19
23	31	43	50	23	34	27	39	16

Fill in this table for Jeffrey's data.

Age	Tally	Frequency
0-19	𝍸	5
20-29	𝍸 𝍸	10
30-39	𝍸	5
40-49	||||	4
50 or over	|||	3 ✓

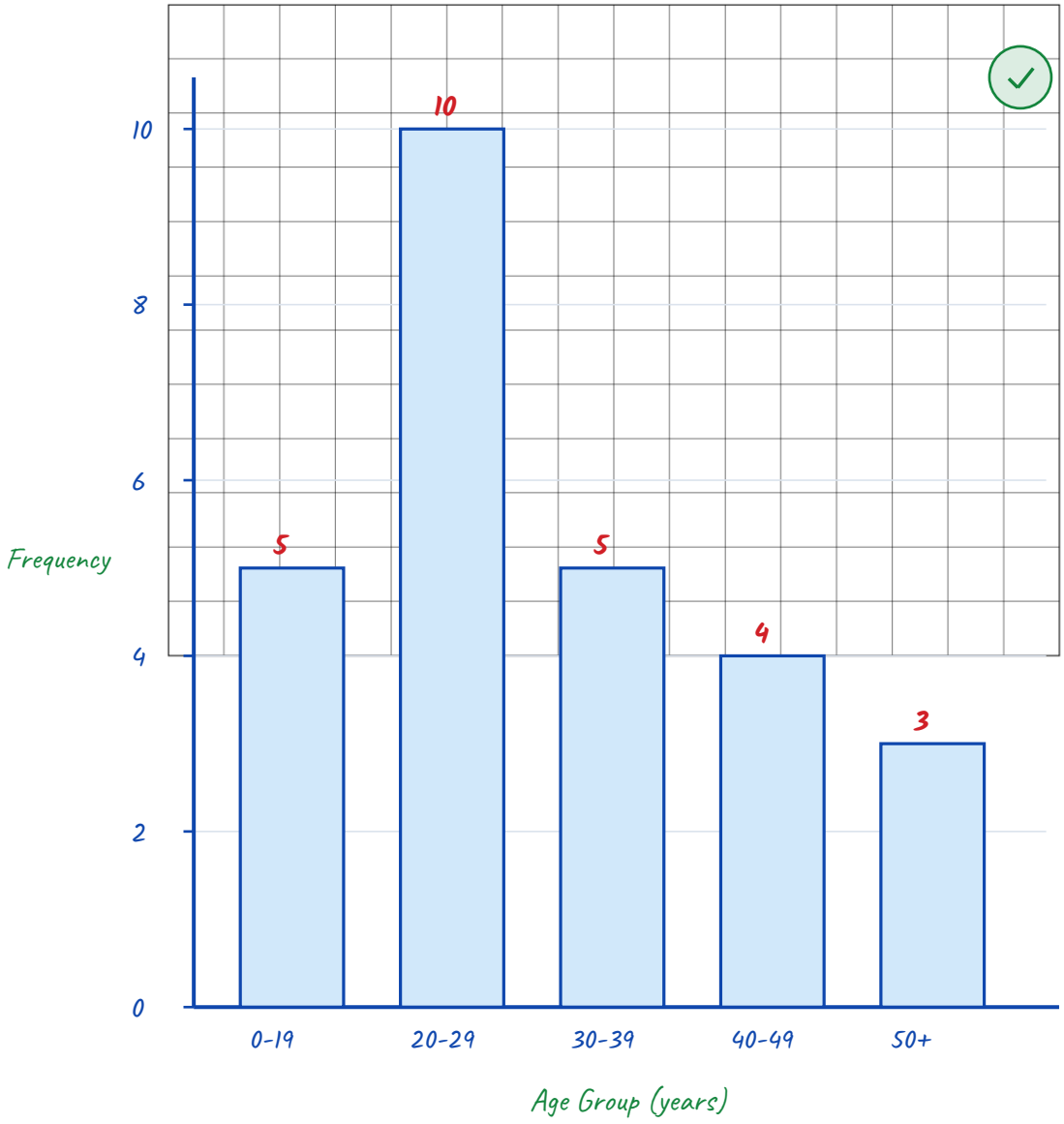
Check total: 5 + 10 + 5 + 4 + 3 = 27 people tallied [2 marks]

Question continues on next page

© https://passfunctionalskills.co.uk/

4 cont. Use this table to plot a bar chart for this data on the graph paper below.

Age Groups of People Attending Jeffrey's Gym **[3 marks]**



[5 marks]

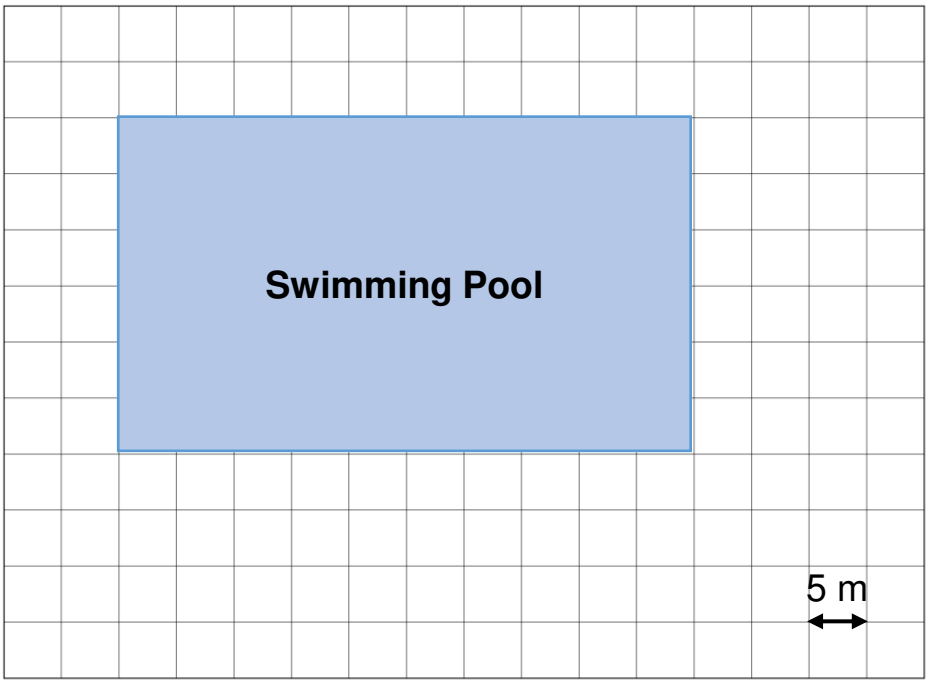
Turn over for next question

5

© https://passfunctionalskills.co.uk/

5 Jeffrey's gym has a swimming pool. Jeffrey draws a scale drawing of the swimming pool on the square grid below.

What is the actual area of the swimming pool?



[3 marks]

Step 1: Pool dimensions on grid = 10 squares long by 6 squares wide

Actual length = 10 × 5 m = 50 m; Actual width = 6 × 5 m = 30 m [M1 M1]

Step 2: Area of rectangular pool = 50 m × 30 m = 1,500 m² [A1]

Answer **1,500 m²** ✓

Turn over for next question

3

© https://passfunctionalskills.co.uk/

6 There are 120 members that attend Jeffrey's gym in total. The gym offer either a silver membership or a gold membership. The ratio of people on the silver membership to gold membership is $3:5$.

How many people are there on each membership?

[2 marks]

Step 1: Total parts = 3 + 5 = 8 parts; 1 part = 120 ÷ 8 = 15 members [M1]

Silver = 3 × 15 = 45 members; Gold = 5 × 15 = 75 members [A1]

Answer **Silver: 45, Gold: 75** ✓

7 Jeffrey has a bag of various sports balls. He has 7 footballs, 3 rugby balls and 6 tennis balls.

What is the probability of Jeffrey picking a football out of the bag at random?

[1 mark]

Total balls in bag = 7 + 3 + 6 = 16 balls

P(football) = 7 footballs out of 16 = 7/16 [1 mark]

Answer **7 / 16** ✓

End of Section A

3

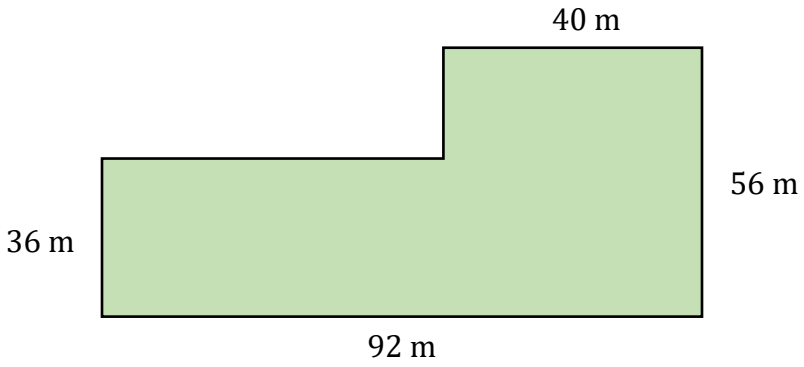
© https://passfunctionalskills.co.uk/

SECTION B : Calculator

Answer ALL questions.
Write your answers in the spaces provided.

© https://passfunctionalskills.co.uk/

8 David is training for a charity run. As part of David's training, he runs around a field seen below. He aims to run at least 5 km.



How many **full** laps of the field will he need to run?

Assume that he can only complete **full** laps.

[3 marks]

Step 1: Missing horizontal side = 100 - 45 = 55 m; vertical = 45 - 20 = 25 m [M1]

Step 2: Perimeter of 1 lap = 100 + 45 + 45 + 25 + 55 + 20 = 290 m [M1]

[M1]

Step 3: 5 km = 5,000 m --> Number of laps = 5,000 ÷ 290 = 17.24 laps

Complete (full) laps run = 17 full laps [A1]

Answer 17 full laps ✓

Turn over for next question

3

© https://passfunctionalskills.co.uk/

9 David records how long it takes him to run 5 km on 5 days in a week. He does this for 3 weeks.

He records his times in the table below.

	Time (minutes)				
	Monday	Tuesday	Wednesday	Thursday	Friday
Week 1	29	32	30	33	32
Week 2	27	29	34	30	28
Week 3	28	30	31	30	33

In which week were the range of his times the greatest?

Show your working.

[2 marks]

Compare range (highest - lowest) for each week:

Week 1: 31.25 - 26.25 = 5.0 mins | Week 2: 29.75 - 27.0 = 2.75 mins [M1]

Week 2 has the smaller range (2.75 < 5.0), so Week 2 is more consistent [A1]

Answer **Week 2** ✓

10 David buys a new pair of running shoes to use for his charity run. The price of the pair of running shoes is £65, but David has a 20% off voucher.

How much will the running shoes cost David?

[1 mark]

20% of £65 = 0.20 × £65 = £13 discount [M1]

Sale price = £65 - £13 = £52.00 [A1]

Answer **£52.00** ✓

Turn over for next question

3

© https://passfunctionalskills.co.uk/

11 There are 24 members of a church. Two thirds of the members are women.

How many of the church members are women?

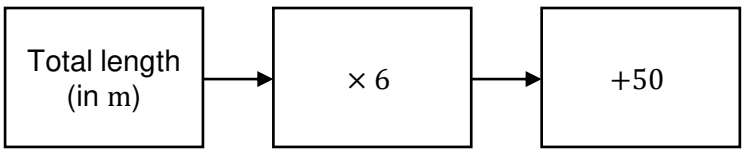
[1 mark]

1/3 of 24 = 24 ÷ 3 = 8 --> 2/3 of 24 = 8 × 2 = 16 women [1 mark]

Answer 16 ✓

12 Delia wants to decorate her square house by putting Christmas lights on two of its sides. Her house is 17 m wide.

To calculate the number of lights she needs, she uses this rule:



Lights come in boxes of 80.

Each box costs £15.99

Calculate how much it would cost Delia to decorate her house in this way.

[4 marks]

Step 1: Area of 1 wall = 4.6 m × 2.5 m = 11.5 m² [M1]

Step 2: Total area of 4 identical walls = 4 × 11.5 m² = 46 m² [M1]

[M1]

Step 3: Tins needed (1 tin covers 12 m²) = 46 ÷ 12 = 3.833... --> 4 tins

Step 4: Total paint cost = 4 × £15.99 = £63.96 [A1]

Answer £63.96 ✓

Turn over for next question

5

© https://passfunctionalskills.co.uk/

13 Shauna is baking cookies for 48 people. She uses the following recipe:

Cookie recipe (serves 6)

225 g butter
125 g caster sugar
300 g plain flour
100 g milk chocolate chips

Shauna has $1\frac{3}{5}$ kg of butter in her fridge.

How much more butter does she need to be able to bake the cookies for 48 people?

Include the units in your answer.

[4 marks]

Step 1: Flour per cookie = 225 g ÷ 15 = 15 g per cookie [M1]

Step 2: Flour needed for 50 cookies = 50 × 15 g = 750 g [M1]

Step 3: Subtract flour Shauna already has (550 g):

Extra flour needed = 750 g − 550 g = 200 g [M1 A1]

Answer **200 g** ✓

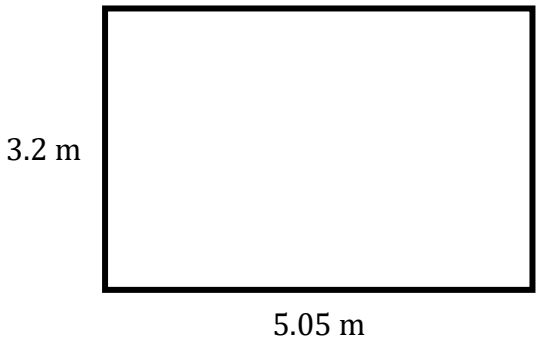
Turn over for next question

4

© https://passfunctionalskills.co.uk/

14 Bob wants to varnish his kitchen floor.

The diagram below shows the dimensions of Bob's kitchen floor.



Bob is going to give the kitchen floor 2 coats of varnish.
1 litre of varnish that Bob buys will cover 5.6 m^2 of floor.
The varnish comes in 2 litre tins, costing £14.95 per tin.

Calculate how much it would cost Bob to varnish his floor.

[5 marks]

Step 1: Kitchen floor area = 5.05 m × 3.2 m = 16.16 m^2 [M1]

Step 2: Total area for 2 coats = 16.16 m^2 × 2 = 32.32 m^2 [M1]

Step 3: Litres needed = 32.32 ÷ 5.6 = 5.771... litres [M1]

Step 4: Number of 2-litre tins = 5.771 ÷ 2 = 2.885... --> 3 tins needed [M1]

Step 5: Total cost = 3 × £14.95 = £44.85 [A1]

Answer £44.85 ✓

Turn over for next question

5

© https://passfunctionalskills.co.uk/

15 The table below shows figures for the number of ticket sales at a museum on one day.

Ticket type	Adults	Children	Concessions	Other
Tickets sold	231	105	56	12

The concession ticket type gives access to the museum to seniors and students. The ratio of seniors to students who visited the museum on this day was $4:3$.

What percentage of the total ticket sales on this day were seniors?

[4 marks]

Step 1: Seniors in 56 concessions (ratio 4 : 3 --> 4 + 3 = 7 parts):

56 ÷ 7 = 8 per part --> Seniors = 4 × 8 = 32 tickets [M1 A1]

Step 2: Total tickets sold = 231 + 105 + 56 + 12 = 404 tickets [M1]

Step 3: Percentage seniors = (32 ÷ 404) × 100 = 7.92% [A1]

Answer 7.92%

Turn over for next question

4

© https://passfunctionalskills.co.uk/

16 Billy weighs 76 kg.

Billy claims that he weighs more than the mean weight of his friends.

The weight of his friends can be seen in the table below.

Friend	Weight (kg)
Kyle	75 kg
Chloe	70 kg
Katie	68 kg
Ari	80 kg
Sean	78 kg
Mark	76 kg

Is Billy correct?

Show your working.

[3 marks]

Step 1: Total weight of 6 friends = 75 + 70 + 68 + 80 + 78 + 76 = 447 kg [M1]

Step 2: Mean weight = 447 kg ÷ 6 = 74.5 kg [M1]

Step 3: Compare: Billy's 76 kg > 74.5 kg, so Billy is correct. [A1]

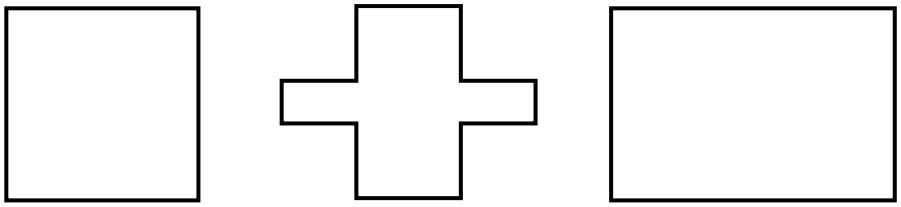
Answer Yes, Billy is correct (76 kg > 74.5 k

Turn over for next question

3

© https://passfunctionalskills.co.uk/

17 Which of these shapes have exactly 2 lines of symmetry?



A B C

[2 marks]

Square A has 4 lines; Rectangle B & Rhombus C each have 2 lines. **[2 marks]**

Answer **B and C** ✓

18 Monique eats $\frac{1}{4}$ of a pizza and Ahmed eats 28% of the same pizza.

Who eats the most pizza?

Show your working.

[2 marks]

Convert Monique's 1/4 to a percentage: 1/4 = 1 ÷ 4 × 100 = 25% **[M1]**

Compare: Ahmed's 28% > Monique's 25% **[A1]**

Answer **Ahmed** ✓

Turn over for next question

4

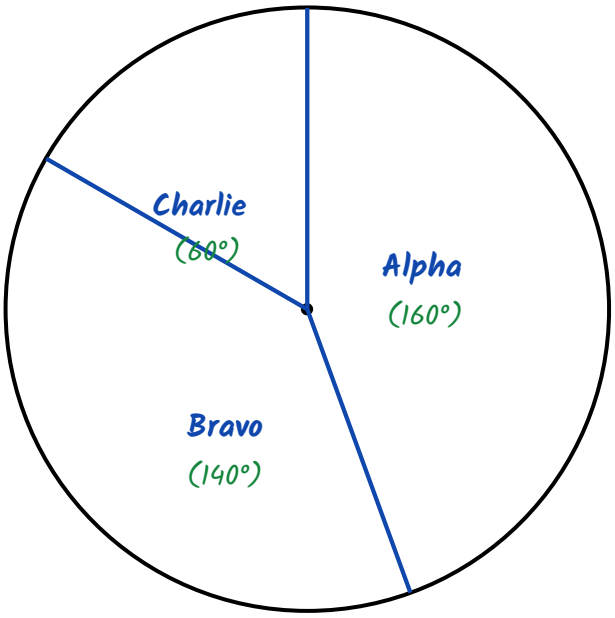
© https://passfunctionalskills.co.uk/

19 Some Army cadets are split up into 3 teams.

The table below shows these results.

Team	Cadets
Alpha	40
Bravo	35
Charlie	15

Complete the pie chart to display this data.



[3 marks]

Step 1: Total cadets = 40 + 35 + 15 = 90 cadets; 360° ÷ 90 = 4° per cadet [M1]

Alpha = 40 × 4° = 160° | Bravo = 35 × 4° = 140° | Charlie = 15 × 4° = 60° [A1 A1]

Answer Alpha 160°, Bravo 140°, Charlie 60°

Turn over for next question

3

© https://passfunctionalskills.co.uk/

20 Mary uses the following mileage chart to see the distances between various cities.

Leicester			
52	**Lincoln**		
26	36	**Nottingham**	
61	47	37	**Sheffield**

Mary wants to drive from Sheffield to Nottingham, and then from Nottingham to Leicester. She has enough petrol in her car to travel 70 miles.

Does she have enough petrol to do these journeys?

Show your working.

[2 marks]

Read distances from mileage chart:

Sheffield to Nottingham = 37 miles; Nottingham to Leicester = 26 miles

Total journey = 37 + 26 = 63 miles (< 70 miles available) [M1 A1]

Answer **Yes (63 miles < 70 miles)** ✓

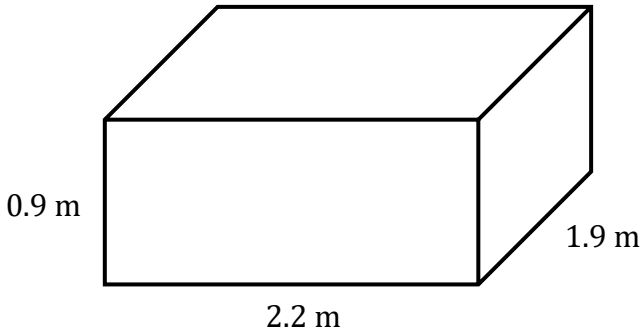
Turn over for next question

2

© https://passfunctionalskills.co.uk/

21 Wendy has bought a hot tub for her garden and wants to fill it with water.

A diagram of the hot tub can be seen below.



She is going to use a hose pipe that will fill the hot tub with $0.026\ \text{m}^3$ of water per minute.

Calculate how long it will take Wendy to fill the hot tub using this hose pipe.

Give your answer to the nearest minute.

[4 marks]

Step 1: Volume of cuboid hot tub = Length × Width × Height:

Volume = 2.2 m × 1.9 m × 0.9 m = 3.762 m³ [M1 A1]

Step 2: Divide volume by fill rate (0.026 m³ per minute):

Time = 3.762 ÷ 0.026 = 144.692... minutes [M1]

Step 3: Round to the nearest minute = 145 minutes [A1]

Answer **145 minutes** ✓

Turn over for next question

4

© https://passfunctionalskills.co.uk/

22 Denise works at a Café and works as a cleaner. Her current working hours are listed in the table below.

Day	Job	Shift times	Breaks
Monday	Café	07:30 – 13:00	30 mins
Tuesday	Café	09:00 – 15:00	30 mins
Wednesday	Day Off		
Thursday	Cleaner	10:00 – 16:00	45 mins
Friday	Café	07:00 – 14:00	45 mins
Saturday	Day Off		
Sunday	Cleaner	11:30 – 14:30	-

Denise is not paid for her break times and she earns £9.10 per hour working at the Café. She earns 20% more per hour working as a cleaner.

If Denise works as a cleaner on a weekend, she gets paid an additional £20.

Work out how much Denise earns in a week.

[5 marks]

Step 1: Paid Café hours (subtracting unpaid breaks):

Mon (5.5 - 0.5 = 5h) + Tue (6 - 0.5 = 5.5h) + Fri (7 - 0.75 = 6.25h) = 16.75 hrs

Café pay = 16.75 × £9.10 = £152.425 [M1 M1]

Step 2: Cleaner rate (+20%) = £9.10 × 1.20 = £10.92 per hour [M1]

Cleaner hours: Thu (6 - 0.75 = 5.25h) + Sun (3h) = 8.25 hrs

Cleaner pay = 8.25 × £10.92 = £90.09 (+ £20 Sunday bonus = £110.09) [M1]

Step 3: Total pay = £152.425 + £110.09 = £262.515 --> £262.52 [A1]

Answer £262.52

5

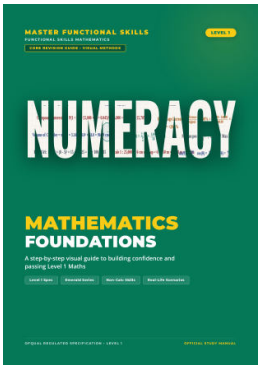
© https://passfunctionalskills.co.uk/

MASTER FUNCTIONAL SKILLS STORE

OFFICIAL REVISION GUIDES, WORKBOOKS & FLASHCARDS

Written by accredited specialist tutors to match the exact 2026 Pearson Edexcel standards

Everything You Need To Revise, Practice & Pass Your Maths Level 1 Exam

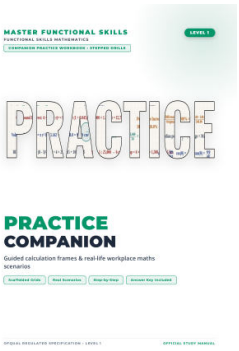


Core Curriculum Manual

Maths Level 1
Core Study Guide

- Full official curriculum coverage
- Section A Non-Calc & Sec B Calc
- Worked models & golden rules
- Dyslexia-friendly visual format

Spiral Bound Print & PDF

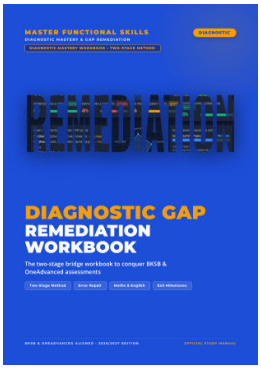


1,000+ Practice Drills

Maths Level 1
Companion Workbook

- 1,000+ exam-style questions
- Progressive difficulty drills
- Full worked step-by-step key
- Targeted fix for weak areas

Spiral Bound Print & PDF



Realistic Mock Papers

Maths Level 1
Official Mock Packs

- Full mock exam paper sets
- Authentic Pearson Edexcel style
- Detailed mark schemes included
- Grade boundaries & pass marks

Exam Booklet Pack & PDF



Pocket Flashcards A6

Maths Level 1
Pocket Revision Cards

- Compact A6 cards for quick study
- Essential formulas & conversions
- Dual-sided question & answer
- High-yield mental arithmetic rules

Boxed Card Deck & Print PDF

Order Official Study Guides & Revision Cards Online

Free Next-Day UK Tracked Shipping · Instant PDF Downloads Included

Trusted by training providers, apprentices & adult learners nationwide.

VISIT THE ONLINE STORE

MasterFunctionalSkills.co.uk